

Combating Cyber Terrorism

A guide to understanding the cyber threat landscape and incident response planning



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RICHARD BINGLEY



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Dedicated to Milena.

Thank you for everything.

ABOUT THE AUTHOR

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His book publications to date include:

- *The Security Consultant's Handbook* (ITGP: 2015);
- *Terrorism: Just the Facts* (Heinemann: 2003); and
- *Arms Trade: Just the Facts* (Heinemann: 2003).

Richard was senior lecturer for security and resilience at Buckinghamshire New University (2012–15) and director of the BNU Business School. He is CEO and principal of the CSARN Global Cyber Academy and a frequent media commentator on cyber security and future technology issues, including recently for the *London Evening Standard* and *Sunday Express*.

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FOREWORD

If you’re anything like me as a reader, the first couple of questions you have before (or after!) purchasing a non-fiction book is *who wrote this and why?* After all, it takes a rather large amount of time for any individual to research, write and arrange the content you find before you.

Probably the first explanation to provide is that I’ve spent more than two decades in a range of job roles that either directly related to undertaking terrorism research or overlapped significantly into having to consider terrorist threat actors as a meaningful risk to one’s own organisations and personnel. During a diverse career, I’ve been an IT manager, governmental spin doctor, cabinet minister events co-ordinator, close protection operative (CPO) and British armed forces member. For more than a decade, I’ve also served as a senior university lecturer and security risk management instructor to a wide array of agencies, military establishments and high-profile business sector clients. Each of these roles has provided me with the privilege to learn and develop from some of the best security management thinkers and organisational leaders out there. I hope that by writing this guide I can contribute something of purpose back to those who work so tirelessly (and often anonymously) behind the scenes to keep us all safe and well.

A couple of quirks of fate also led me to write this book and focus on the specific content that I did. I narrowly missed the 7 July 2005 Al-Qaeda London transport bombings. That morning, heading into work in Westminster (planning to take a half-day as it was my thirtieth birthday), I fortuitously ran late into Liverpool Street station. I missed the Circle Line tube explosion near Aldgate by four or five minutes. Many years later, as Plymouth City Council leader, I was part-responsible for the city’s crisis recovery programme, put in place after spree killings conducted by an irate incel chat forum user. These and other life

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events sometimes force us to think existentially as well as plan much better to ‘expect the unexpected’. Also, to corral those around us into action! I don’t think that I’m unusual in being ever-so-slightly impacted by perpetrators of terrorism. And I’ve certainly never been physically harmed, unlike so many others less fortunate. Nonetheless, attempting to prevent terrorism, or helping others to counter it, has been a consistent theme in much of my work. This book, I guess, is the product of that desire to raise awareness and offer guidance in relation to risk management and incident response.

I originally arrived at the idea for producing a ‘cyber terrorism’ manuscript after feeling – despite a complete lack of science to back my hunch up – that many contemporary terror cases bore so much dependency upon Internet-enabled computer systems. Social media and the roll-out of interactive web 2.0 platforms since the millennium became ubiquitous, not just in the US and wealthier economies. Several countries moved swiftly to wrap controls around public use and placed direct responsibility for content transmission squarely onto the shoulders of Internet service providers. Most others didn’t – and, for better or for worse – it’s this unregulated legacy, fifty years after the Internet’s inception, that so many countries and companies are challenged by today.

When social media spawned effectively, offering unvetted end users 24/7 availability and functionality, a glorious ‘genie’ escaped from bottle, for those considering active political extremism. Violent non-state actors – no matter how geographically remote, politically insignificant, or ideologically warped – could now easily interact with audiences well beyond their hitherto limited theatres of operation. Fringe figures from anywhere, prone to promoting violent discrimination and sectarianism – often shunned by civil society within their home jurisdictions – now had prime access to an intoxicatingly reliable instrument of international publicity and bilateral audience engagement.

Better still (for the terrorist), picture-driven content could be tailored for, and literally hand-delivered directly to, digitally-addicted mass audiences. Violent extremists conducting the most awful atrocities found that they could bypass mainstream news organisations and set viewing agendas themselves. At the same time, mainstream media editors were effectively pressurised to cover the perpetrator’s grisly insider view. If some news agencies tried to do the right thing by not always following the maxim ‘if it bleeds, it leads’ and not offering the low-end clickbait, they risked sinking into obscurity and riling their shareholders.

For extremist orators and terror recruiters, cleverly-worded manipulations, defamations and damnations began appearing routinely on our smartphone and tablet screens. These home-brewed radical news feeds, packaged reportage and scripted comment with catchy soundbites, arrived just as quickly, authoritatively and slickly as productions broadcast by mainstream news anchors and documentary makers. Advanced tech certainly ushered in ‘information democratisation’. But alongside that gain sat a significant degree of anarchy too. It is rather paradoxical, it seems, that as much of the world came together to better coordinate international counterterrorism after the 9/11 atrocities in America, such obvious human security challenges posed by information communication technology (ICT) advance were left almost entirely unaddressed in the US, EU and beyond.

This book will demonstrate that cyber terrorism gained significant – some might say, unhampered – traction from the early 2000s. The notion of ‘cyber terrorism’ became of primary importance in understanding so much of the necessary detail that really lies within the phenomenon of contemporary terrorism. Nowadays, because of widespread access to advanced communications and robot technologies, terrorism activities are so much more diffuse, randomised, unpredictable. And, one might plausibly argue, unaccountable. (Some terror groups and politically extremist organisations appear to have lost complete centralised control of operations conducted under their banner.) Such evidence pointing to the digitally-enabled decentralisation of violent extremism has recently been borne out in many formal investigations, inquests and judicial reports. Computer forensics evidence presented by prosecutors in court demonstrate that digital media platforms have often become the *single most important reason* why an individual ‘activated’ and moved forward along an emotional continuum towards carrying out a physical terrorist attack.

Since the early 1800s, a panoply of terrorism scholars, authoritative news organisations and governmental institutions have come to define terrorism as pre-meditated political violence carried out by *non-state actors*. (Although, the term hailed from the bloodthirsty actions taken during 1793/4 by France’s post-revolutionary ‘government’: the *Reign of Terror* or *Règne de la Terreur*.) Confusion has reigned a little when it comes to discussing ‘cyber terrorism’. When I scoped out writing this book, I couldn’t understand why so many news references to cyber terrorism were not applying this elegantly simple *non-state* definitional criteria. Especially because, as I felt, so much cyber terrorism is being perpetrated by non-state actors; many of whom are distinctly *anti-government* in worldview.

Indeed, the biggest difficulty I faced in writing about cyber terrorism was in deciding which cases and aspects to omit and why. For the reader’s benefit, I wished to safeguard twin goals: keep it brief and deliver an overview. This book could easily have been three or four times the size. Examples and case studies presented are only a snippet of what’s going on out there.

Thus, this book is principally an overview and guide, laced with case studies and useful signposting. I hope it prompts further, much more detailed, research. (Please get in touch if you take up this challenge!)

Therefore, this book does not *per se* cover state-sponsored cyber terrorism. Namely, cyber attacks or the facilitation of physical attacks either directly authorised and implemented by government or by a ‘plausibly deniable’ proxy. There are a couple of exceptional case studies and references where the chain of responsibility is sufficiently hazy, or attack methods have been shown to be of documented interest to terror groups. However, most threat intelligence analysts would likely understand that government-directed or sanctioned cyber attacks – designed to terrorise or kill human targets – tend to fall within the *cyber warfare* or *warfare* definitional domain. I leave that very fertile and expansive research territory to be harvested by other authors.

Following the COVID-19 Pandemic, a drive to hybrid working, the physical fall of ISIS on the battlefield, and the 2021 Capitol Hill riots, still no dedicated cyber terrorism guide for business appeared readily available. Security risk managers and chief information security officers (CISOs) also still eagerly awaited the publication of peer-reviewed industry standards that could shed light on the best approaches to identify and mitigate the security risks posed by advanced technology. For example, an artificial intelligence (AI) security risk management standard was not due out until 2025. In this book – as part of addressing imminent advanced tech cyber terrorism concerns – we therefore describe and excavate tools and products found in the domains of AI, chatbots, drones and cryptocurrency. We explain how these powerful capabilities are being exploited by terrorist entities. We explore risk management frameworks that might be quite traditional but explain how they can be applied to our contemporary, advanced digital ICT world.

A personal note now. This is my fourth published book. It has been, by far, the most difficult to complete. In part, because the target topic is so potentially vast and the exploitation possibilities within the cybersphere are characterised by a level of speed and complexity that few human beings can easily cope with! The strain of writing any book is physical and mental. It’s a long, drawn-out, emotionally grinding process. You spend a lot of time working alone to maintain focus and grip the issue at hand. This uber-focused psychological mode soon begins to feel like punishing austerity. A little like voluntarily locking your mind into solitary confinement for a year or longer. Outside of your day job (mine’s teaching), you can’t read much beyond related books and articles. (Many are referenced throughout and at the end: approximately 360 in total.) Exciting new books and films pass by unobserved. Professional frustration lurks beneath the surface too. News reports seemingly break every day, related to ‘your topic’. But if you move to comment upon, or study them in any depth, you have ultimately become distracted...prolonging the torment of failing to finish that manuscript! Then arrive various stages of review and edit. But we (and I say ‘we’ because it’s not just me) got there in the end.

I would therefore like to record very important ‘thank yous’. Firstly, to my publisher, ITGP, and Publications Manager, Nicola Day, for her Confucian patience during the writing process. I would like to thank Kirsty Ridge, Copy Editor at GRC International Group PLC, for her help copy editing the material in this book. I completely underestimated the size and scale of research and content required. I’d also like to thank the book’s two reviewers. Firstly, Simon King, formerly senior lecturer in security and resilience at Buckinghamshire New University; a good colleague back in our university teaching days. I’m pleased to say that nowadays we both discuss cricket far more than security dilemmas. His forensic feedback massively enhanced the first draft. Our second reviewer, Ze’ev Portner, serves as a university lecturer in law. Ze’ev possesses a rich pre-academia career background having worked at the Houses of Parliament as a chief of staff. Ze’ev was also employed at various stadia within the UK security industry. He started his legal career with law firms in Israel and London. Ze’ev’s positivity about this book’s relevancy for non-technical readers really helped shape our approach to structuring and fine-tuning the finished product.

Thanks also to Marshall Kent, former counterterrorism coordinator at London’s Metropolitan Police. Marshall kindly granted an interview and I’ve also found our recent conversations really insightful and valuable in pushing cyber terrorism considerations forward to officials and policymakers. ‘Thank you’ similarly to the following: Helen Prendergast, who so ably and patiently supported me in my *other world* during 2022/23. David Evans (founder and chairman) and Andy Williams (deputy chair) at the Global Terrorism Information Network (TINYg); in my view the best public-private sector partnership for counter-terrorism discussions out there. The late Graeme McGowan – formerly at GCHQ and the Home Office – for his inspiring and tireless support, energy and positivity. (GMG sadly passed away shortly before this book’s publication.) Finally, much love and thanks to my wife, Milena, for her consistent support and belief in me.

It seemed to me by 2021 – after two decades in and around the security world – that all my conference talks, course writing, teaching and pub-bar warnings, counted for zero...unless I could successfully encapsulate the issue at hand: cyber terrorism. Then, explain coherently, the critical importance of preventing some of its devastating impacts. For me personally, it was time to ‘put up or shut up’. To bring the security sector’s thoughts, research outputs and media commentary together. As Arnold Schwarzenegger famously said: “*You can have results or excuses.*” I’ve tried here to deliver a basic overview and snapshot

of cyber terrorism. Anything beyond that – such as our readers’ updating their security plans, or recommending this ITGP book to others – would clearly be a positive result!

Richard Bingley.

London, 2024.

CHAPTER 1: INTRODUCTION

Security practitioners are increasingly confronted with a chilling phrase: *cyber terrorism*. For many, such as me, it conveys a sense of fear and hopelessness at the same time. What is this thing called ‘cyber terrorism’? Moreover, what can we begin to do about it? Until it hits us, the impact feels a million miles away. Another person’s nightmare. Another organisation’s problem. Another country’s war.

For the purposes of precisely understanding cyber terrorism throughout this book, we have borrowed and applied the National Cyber Security Centre’s core two principles used to define and characterise a cyber crime:

- 1. Cyber-dependent crimes** – crimes that can be committed only through the use of information and communications technology (ICT) devices, where the devices are both the tool for committing the crime, and the target of the crime.
- 2. Cyber-enabled crimes** – traditional crimes that can be increased in scale or reach by the use of computers, computer networks or other forms of ICT. (CPS: 2019)

We will demonstrate how terrorists intentionally use computer systems to attack and harm human beings and property. They also use computer systems to diversify, complicate and increase terrorist attack impacts, including body counts and escalated public panic. Or to provide **disinformation** and **misinformation** to target audiences, including the emergency services and/or investigators, to sabotage aid, medical assistance and evidence. Nefarious ICT users, programmers and programs (including much AI-powered software) have all been instrumental in recruiting, inspiring, training, executing and amplifying acts of terrorism, which has resulted in the loss of life and/or lifechanging physical injuries that could never have occurred without cybersphere support and facilitation, whether by inanimate technology or others on the network using it. All of the above can be encapsulated by the phrase ‘cyber terrorism’.

The issue at hand

The Internet is an integral part of everyday life for the vast majority of businesses and individuals. According to the Department for Digital, Culture, Media & Sport, even before the homeworking revolution prompted by the COVID-19 pandemic, nine out of ten adults live (and sometimes breathe) our lives online. We use the Internet in many ways to enrich and transform our lives. For instance, billions of us use the Internet for free lessons for leisure, formal education or to improve our professional prospects. But here’s the downside: *“In the wrong hands the internet can be used to spread terrorist and other illegal or harmful content”* (DCMS: 2019).

This government warning is particularly foreboding given that access to the Internet is becoming less voluntary. Web access has become viewed as an existential human right, a pre-requisite of citizenship and societal belonging. In some jurisdictions, and during some public safety incident scenarios (such as pandemic management and military curfews), using online public services is the *mandated* method for contacting public sector support and accessing critical national infrastructure. (A lot of companies and public bodies have completely junked customer telephone lines and other routes to informal support.) Ergo, like it or not, it seems we must all be online.

This state of play is not just applicable to adults who are supposedly more life-experienced and threat-savvy. In most advanced industrial economies, 99% of 12 to 15 year olds regularly use the Internet too. In most cases, children travel about cyberspace with relatively unfettered limitations. And just like their elders, they’ve experienced little if any cyber security awareness training. The flip side to such joyous online libertarianism is now becoming apparent. According to British and European police forces, we are witnessing record numbers of children radicalised and prosecuted for online extremism (Counter

Terrorism Policing: 2021). During chapters 2 and 3 and elsewhere – using some cases in point – we confront cyber security vulnerabilities related to digital netizens who are underage or in their early teens. Similar to adults who experience behavioural mental illnesses, underdeveloped younger brains are at an additional risk from becoming attracted to and sucked into an online whirlpool of physical self-harm as well as public disorder and violence.

Despite well-meaning interventions by a range of influential stakeholders (tech firms, governments, police and academia alike), our computer networks remain riddled with easily-accessible horror images, violence-inspiring proclamations and instructional *how to* videos guiding the weak and angry into executing acts of terrorism. For billions of end users, such vivid, violent material is plastered over social networking sites by way of pop-ups, short URLs and preview videos. Social media forums, including gaming forums, boast monthly active audience sizes that are comparable to the world's most populous nation states or continents.

Therefore, accessing terrorism content does not require much in the way of research skills, technical ability or patience. AI-powered search engine prompts, hashtag funnels and slick terrorist-operated digital marketing strategies fuse together, preying upon end users and hooking into those most psychologically receptive. Out of the many millions who might catch a glimpse of such a pop-up image, or click through into an extremist website, either in error or innocent curiosity, it only takes a tiny fraction to become hooked for us all to have a major problem. Because a tiny fraction of a million views translates into a new legion of recruits to yet another terror network. In chapters 2 and 3, we learn how it is not just Al-Qaeda and ISIS that attempt to stimulate and nurture a 'cyber caliphate'. But that emerging terror networks across Europe and America – including the extreme far right and affiliated teenage hacktivists – are conducting copycat operations and have come to define themselves proudly as 'generation terror'.

According to Tech Against Terrorism (TAT), the non-governmental organisation (NGO) formed by big tech firms to help counter terrorism, the organisation tracked 45,000 URLs from 2016 to 2019 that led to numerous websites hosting terrorist material. This number was just a snapshot (read, massive underestimation), the group felt. Some 330 digital media platforms were used by terror groups for broadcasting, communications and file sharing. Of these, half were small or micro-platforms (Tech Against Terrorism: 2019). Nonetheless, during the span of this book's research (2022–24), the preponderance of terrorist material with mass reach, and arguably greater impact, was hosted on the super-sized social media platforms. Therefore, throughout chapter 3, we closely examine the role of several of the largest social media platforms. We also demonstrate and affirm TAT's findings that terror activists and material are fast moving onto the anonymously run decentralised web (DWeb). Although, because the DWeb reaches far lower audiences, terror groups still tend to use the pre-eminent digital platforms for signature event communications while retaining and deploying DWeb forums for niche discussions and/or for business continuity backup solutions following various waves of government/tech-industry takedowns.

Meanwhile, for those more technically adept, the DWeb (which we cover at the back-end of chapter 3) offers a technical playground where warfare fantasies and dark operations can be enjoyed. It remains relatively easy for tech-savvy netizens to quickly become an integral part of online terror communities, including the push for #online jihad, a hashtag phrase increasingly used beyond ISIS and Al-Qaeda domains. The thrill and attraction for programmers and software developers in carrying out core technical tasks on behalf of a terror organisation and its ringleaders, and building bespoke platforms (mainly for gaming and training), is observable across countless decentralised web programming forums. At the lower technical level, tasks for online terror supporters tend to include collaborative assistance in helping to build **click farms** and fake social media accounts. Meanwhile, more advanced technical assistance can involve shared project work between programmers and software engineers to design and distribute malware for direct cyber attacks. Or more subtle requirements to help with technical eavesdropping and technical surveillance (including spyware injections and development), **honeypot** exploits, or even enticing the enemy into physical or online honeypots. Most tech support terrorists work from home, commit to office hours, have a boss, follow orders and collaborate online. Hybrid working at its most malignant.

As this book came to be researched, terrorism causation data began to noticeably shift. Although we should remain circumspect and not pre-judge pattern analysis in mid-flight, there appears to have been a tangible increase in politically motivated terror incidents within North America, Europe and Oceania. According to the Global Terrorism Index (GTI), politically motivated terrorist attacks came to outnumber religious-inspired attacks by the early 2020s (Global Terrorism Index: 2022). If this is correct, it appears to be that the pendulum of terrorism activity within most 'western hemisphere' political environments has swung back at least slightly towards violent political extremism related more to policy rejection than religious mission.

Data from Europol and European police agencies also shows a clear uptick in extreme far right terrorism activity across much of the Western hemisphere. With a sense of foreboding, this book's research points towards politically motivated domestic terrorists – including violently radicalised children – being shepherded into terrorism by the same cyber-enabled features that stoked and sustained two decades (and counting) of jihadist attacks. Perhaps a central lesson that all terror groups have learned

from ISIS is best summarised by the Washington Post's **Joby Warwick**: *"In the [shrunk] life cycle of news — normal, breaking news — things tend to disappear quickly anyway [...] If the Islamic State can extend the news cycle to 30 hours before being shut down, they've succeeded"* (Washington Post: 2018).

Furthermore, due to Web 2.0's democratisation of content production, which empowers any end user to broadcast video material, terrorists can circumvent the media almost entirely. According to Professor Hagai Segal at New York University: *"Today anybody can be a journalist [...] Terrorists are journalists and they recognise their capacity to generate news but also to drive the news agenda [...] in a way that is designed to create that psychological impact."* Segal cites the three days of terror attacks in Mumbai, India, during 2008, as a pivotal news event whereby terrorists came to realise that by attacking civilians in key tourist centres, they could also significantly influence the lives of massive video audiences; viewers who found themselves unable to switch off from the sequence of atrocities. (Vimeo: 2015). In his paper *ISIS and Innovative Propaganda*, digital media expert Dylan Gerstel observes: *"ISIS propaganda in the West targets 16-25 year olds who are isolated from their societies and who do not have a strong sense of identity or purpose"* (Gerstel: 2016).

Several years later, and much depleted by military defeat in Syria and much of Iraq, Washington-based terrorism monitoring service, MEMRI, argues that *"ISIS media dissemination [is the] most important means of support"* (MEMRI: 2022b). It seems that such vividly successful cyber warfare lessons – which directly transformed ISIS's recruitment strength way beyond Al-Qaeda's limited reach – has now been researched and processed by other violent extremist groups. What this means for public safety is beyond this book's scope. Nevertheless, throughout **chapter 2**, we examine the dominant cyber terrorism ideologies or lines of thinking (idealisations). The book will reflect upon several cases in point from each ideological categorisation.

In **chapter 4**, we look at the use of cyber terrorism against industry and infrastructure while also considering potential exploits of advanced technologies. A little more than a decade ago, former CIA Director and US Secretary of Defense, Leon Panetta, suggested that the world was *"facing the possibility of a 'cyber-Pearl Harbor'"* that could *"dismantle the nation's power grid, transportation system, financial networks and government"* (Bumiller and Shanker: 2012). While, to our knowledge, this near-apocalyptic threat hasn't quite materialised to date, we examine how, *inter alia*, AI, chatbots, cryptocurrency, drones and GPS eavesdropping and sabotage can become (or may already have become) weaponised tools within the modern terrorist armoury.

As we confront the terrorists' use of tools, tactics, ideologies and responses within cyber-enabled domains, it is refreshing to note that this book has a rare advantage over most other security-dilemma-related fact books. Namely, the burden of proof as to whether tech firms and policymakers need to deliver far more tangible outcomes in combatting cyber terrorism seems already to have been proven beyond all reasonable doubt within the court of public opinion. Opinion polls across America found that the public views cyber terrorism as the third-most critical threat over the next decade – ranked ahead of fears about China's military power or conflict in the Middle East (McCarthy 2016; Norman 2018). Consistently throughout this book, we provide commentary and guidance as to the nature of the threat and how individuals, families and businesses can recognise the symptoms and do something about them! **Chapter 5** – our final chapter – is therefore dedicated to security planning and incident response.

Finally, we point you to this book's closing pages that provide an alphabetical list of 'Key organisations and digital platforms' cited, an alphabetical list of and 'Description of terrorist organisations' cited, links to online summaries of terrorist attacks, and a glossary of terms used within this publication. We also provide Appendix E – kindly reproduced from the UK Crown Prosecution Service – that defines and characterises each UK cyber crime and legislation that it can be prosecuted under. Some key organisations, platforms and groups described are highlighted in bold at the first point of reference when reading through this publication.

1.1 Definitions and criteria

At the time of writing, there is no universally accepted definition of 'cyber terrorism'. Determining a definition for terrorism has been problematic for law enforcement agencies, policymakers and academics alike. According to the US **FBI**, cyber terrorism is any *"premeditated, politically motivated attack against information, computer systems, computer programs, and data, which results in violence against non-combatant targets by subnational groups or clandestine agents"* (Dataconomy.com: 2022). A concrete definition is also provided by a local authority in the UK. Wigan Council's cyber team summarises cyber terrorism – also known as 'digital terrorism' – as: *"Disruptive attacks by recognised terrorist organisations against computer systems with the intent of generating alarm, panic, or the physical disruption of the information system."* The council goes on to identify several potential methods of attack that can be fairly construed as cyber terrorism: *"The internet can be used by terrorists to finance their operations, train other terrorists, and plan terror attacks. The more*

mainstream idea of cyber terrorism is the hacking of government or private servers to access sensitive information or even siphon funds for use in terror activities” (Wigan Council: 2023).

The UK Terrorism Act 2000 defines terrorism as the use or threat of one or more of the actions bullet-pointed below: Specific actions included are:

- Serious violence against a person;
- Serious damage to property;
- Endangering a person’s life (other than that of the person committing the action);
- Creating a serious risk to the health or safety of the public or a section of the public; and
- Action designed to seriously interfere with or seriously disrupt an electronic system.

The Act supposes that these actions are designed to influence the government, or an international governmental organisation, or to intimidate the public. For terrorism to have occurred, the action or threat must be for the purpose of advancing a political, religious, racial or ideological cause. Actions or threats that become investigated and prosecuted can occur outside the UK.

The use or threat of action, as set out above, which involves the use of firearms or explosives, is terrorism regardless of whether the action is designed to influence the government or an international governmental organisation, or to intimidate the public or a section of the public (CPS: 2023). The Act also extends into support and planning terrorist activities. According to terrorism guidance issued publicly by the Crown Prosecution Service: *“It is important to note that, in order to be convicted of a terrorism offence, a person doesn’t actually have to commit what could be considered a terrorist attack. Planning, assisting and even collecting information on how to commit terrorist acts are all crimes under UK terrorism legislation”* (CPS: 2023). Given the complexity, criticality and heavy-duty security requirements involved in processing (or not) terrorism prosecutions, the CPS operates a dedicated Special Crime and Counter Terrorism Division (SCCTD). This department provides *“early investigative advice to the police, make charging decisions and prosecute individuals accused of committing an offence”* (CPS: 2023).

Beyond the UK, (although London is a key member) a grouping of 57 nations form the Organization for Security and Cooperation in Europe (OSCE). This international body conducts and supports an impressive range of conflict prevention and counter-extremism work across its purview territories straddling North America, Europe and Central Asia. Several years ago, following mass casualty terror attacks inside the US, UK, Europe and Russia, the OSCE identified a “typology of terrorism”; highly useful because so many countries could not agree upon a definition and therefore formal international partnership working and guidance development was tricky. The OSCE’s timely typology drew together many common and agreed strands from each country that define an act of terrorism. The typology that an act of terrorism was:

- Of an organised nature, whether the organisation involved is large or small;
- A danger to human life, limb or property;
- An attempt to intentionally undermine democratic government. In particular, by trying to influence policy and lawmakers; and
- Characterised by randomness and consequential spreading of fear/terror among a population (OSCE and ODIHR: 2007).

Government analysts and practitioners across the UK and US have coalesced around asserting four key indicators within the planning stage that point towards defining a major security incident as a “terrorist” event:

1. Planned beforehand – in ‘cold blood’.
2. Carried out to achieve political change.
3. Aimed at non-combatants/civilians.
4. Usually the work of a group (not necessarily a group setting).

(Source: CIA’s Paul Pillar cited in Bingley (2003).)

Leading academic scholars within terrorism domains appreciate strict definitions, typologies and categorisation because it enables them to conduct research within clear parameters. Terrorism definitions from such scholars are also very useful because they bypass internal diplomatic sensitivities and governmental definitional preferences, which can often be perceived as biased within territories suffering from civil war or polarised social and ethnic conflicts. Therefore, identifying a definition

of ‘terrorism’ acceptable to most sides and parties may be too complex or lengthy an issue to revolve. For this reason, we now include a terrorism definition provided by leading academic experts, including psychiatrist **Jerrold Post**:

“Premeditated, politically motivated violence – or the threat of violence – against noncombatants or property by subnational groups or clandestine agents to influence, coerce, or intimidate an audience extending beyond the immediate target of the attack”

(Post, Ruby & Shaw, 2000).

The leading researcher behind criminal aggression theorisation as well as group identity and dynamics, **Albert Bandura**, states: “*Terrorism is usually defined as a strategy of violence designed to promote desired outcomes by instilling fear in the public at large*” (Bandura: 162). Award-winning US journalist and author **Ron Suskind** wrote in his insightful book *The One Per Cent Doctrine: Deep Inside America’s Pursuit of its Enemies Since 9/11* that: “*People will do all sorts of things to make fear stop – so creating fear is a goal of the terrorist*” (Suskind: 2006). Such motivation by some individuals and groups to coerce and control large target audiences, in essence, partly explains why cyber terrorism spawned so speedily and omnipotently across very many digital media platforms during the tech sector’s somewhat unregulated formative years. But this conundrum also takes national policymakers and advisers (and, indeed, organisational leaders) into challenging intellectual territory. Namely, to what extent can they or should they seek to intervene in, or even impede, information communication technologies that function (as neutral data package carriers) across their jurisdictions? In many states, terrorism incitement and glorification offences have been instituted that typically require a substantial burden of proof before prosecutors present a case before court. In turn, jurors and/or judges should only convict upon the evidence being presented as being so compelling that the defendant’s guilt is factual “beyond all reasonable doubt”. Such offences – primarily based upon the defendant’s use of communication tools and techniques – are incredibly difficult to fairly prove and, therefore, rarely successful. It’s worth recalling that the introduction of such laws were strongly resisted by policymakers beforehand, due principally to fears that law enforcement agencies would become enmeshed in policing ‘thought’ or that those who were radicalised (or suffering mental health difficulties), but whom were non-violent, could be prosecuted or imprisoned. Nevertheless, democratic societies – particularly across Europe – have broadly accepted that incitement and glorification statutes within counter-terror legislation have had to be introduced and/or strengthened.

However, the perceived spread of cyber terrorism has now pushed policymakers, tech firm chiefs, and other organisational leaders, into discussing levels of ICT intrusion that potentially move the scales further towards significant mass monitoring and intervention capabilities. A hypothesis emerging within counter-terrorism analysis fields called ‘Stochastic terrorism’ refers to influential media figures, including political leaders, that publicly demonise or vilify others in such a forceful manner that acts of political violence could be inspired against the target within the speech. Such language and rhetoric may not cross the legal threshold to be considered in breach of incitement laws. However, techniques could include the use of demeaning personalised jokes, coded or vague references, or a crude ‘dog whistle’ remark (for example, naming a sole individual as responsible for an unpopular political outcome). Such rhetoric, whereby the instigator can plausibly deny responsibility for any tangible outcome, would fall short of incitement laws in many jurisdictions. For example, in the UK and US, legal incitement is generally understood to mean speech that is intended to produce “*imminent lawless action*” (Amman and Meloy: 2022).

1.2 Laws and Regulations

UK

The **Terrorism Act of 2006** established offences that in effect relate to the government’s ongoing ‘Prevent’ terrorism strategy. They include, in particular, the offence of encouraging terrorism or disseminating publications that seek to encourage political violence. Such offences of incitement to terrorism are sometimes known as *glorification* offences. The legal framework for this provision is provided by Sections 1 and 2 of the Terrorism Act 2006 (TACT), which create the offences of encouragement of terrorism (s.1) and the dissemination of terrorist publications (s.2). Section 3 of TACT provides that those served with notices who fail to remove, without reasonable excuse, the material that is unlawful and terrorism-related within a specified period, are treated as endorsing it. (HMG Prevent Strategy: 2011: 77).

The **Counter-Terrorism and Security Act 2015** determines that schools and universities must help divert students from recruitment into terrorist groups, as part of Prevent. Educators are expected to report students suspected of terrorist or extremist sympathies to a local government body, and vet the remarks of visiting speakers, among other obligations (Freedom House: 2022).

The **Counter-Terrorism and Border Security Act**, which received royal assent in 2019, included provisions related to online behaviour. Individuals can receive prison sentences of up to 15 years for viewing and accessing material that is useful or likely to be useful in preparing or committing a terrorist act. The law provides for exceptions for journalists or academics who access such materials in the course of their work but does not sufficiently explain other possible circumstances in which access might be legitimate (Freedom House: 2023).

The **Counter-Terrorism and Sentencing Act (2021)** has made provisions for jail sentences of up to 14 years for anyone who “*supports a proscribed terrorist organisation*” (HMG Legislation: 2023). In addition, the Act materially ends the likely prospect of early release for anyone convicted of a serious terror offence and forces them to spend their whole term in jail. The maximum penalty for membership of a proscribed terror organisation increased from 10 to 14 years under this Act (GOV.UK: 2021).

A proposed **Online Safety Bill**, initially tabled before the Houses of Parliament during 2021, still had not passed all the legislative committee stages at the time of writing. Scuppered by internal political changes within the government, as well as uncertainty as to the level of content intervention expected of Internet service providers (ISPs), the bill’s proposals are sanctions-based. They include giving power to Ofcom, the regulator, to fine online services the greater of £18 million or 10% of global turnover, if they do not comply with the bill’s provisions. According to some reports, social media giants would be required to quickly remove illegal content such as hate speech or “*face hefty penalties and even potential criminal prosecutions for tech executives who fail to act*” (Politico: 2023).

Draft Terrorism (Protection of Premises) Bill: This proposed law is also known as ‘Martyn’s Law’ in memory of Martyn Hett, one of 22 people killed in the 2017 Manchester Arena bombing attack conducted by a Manchester-born **Al-Qaeda** operative. The subsequent formal inquest by **Sir John Saunders** found that several key aspects of security planning, threat identification and incident response failed before, during and after the attack. The proposed bill focuses mainly on mandating security risk assessments and implementing basic security controls at publicly accessible locations and/or events hosting more than one hundred people. The bill was lightly criticised by the Home Affairs Select Committee in July 2023 for proposing requirements that could potentially over-burden businesses, while not actually preventing further mass casualty attacks. Whatever happens to this piece of legislation, the lessons for cyber security risk managers are clear. Data owners should be aware that the secure ownership, storage and distribution of personal data is a fundamental counter-terrorism requirement (see CCTV case study later). Likewise, sensitive site security documents (including site, evacuation incident response, crisis management and training plans) must be kept securely and safely away from access by would-be terrorists or witting and unwitting accomplices. Employee naivety in relation to publishing very useful advance information, including site photos, staff and planning details – is a recurring feature running throughout so many post-incident reports into terrorism. Such data breaches and privacy indiscretions could come with real legal sanctions if Martyn’s Law or other pieces of future counter-terrorism legislation ever catch up with the cybersphere.

Case study: CCTV, data storage and the law

Proposed new counter-terrorism legislation potentially enacted by the aforementioned Draft Terrorism (Protection of Premises) Bill is likely to enshrine a *Protect Duty* whereby organisations must better plan for and secure their locations. Within this law, organisational incident response policies and practices, including expectations of training a cadre of counter-terrorism ‘competent persons’, are all expected to become mandated. This proposed statute and variations of it are being closely looked at by many other jurisdictions around the world. (Even if the UK Premises Bill stalls, the following good practice can be considered *the right thing to do*, as well as very much legally advisable!)

One of the key challenges of countering cyber terrorism is understanding precisely how laws impacting the physical security space will be implemented and operationalised, within any business. Genetec’s Steve Green observes that gathering most digital evidence, including CCTV images, was hitherto largely dependent upon “*manual processes such as sharing footage on USB sticks or burning it to DVDs*”. Green advises that “*the Cloud offers much better options for a faster, safer and more accountable means of providing access to digital evidence*” (City Security Magazine: 2023: 9). According to the National Counter Terrorism Office (NACTSO) guidance on Publicly Accessible Locations, “*If an incident occurs, the [CCTV] images should be copied onto removable media. A record should be kept of movement of the CD or CDR disk, SD card, or other media showing an identification number, times, dates and names of those handling the media. All used media should be destroyed or disposed of securely*” (PALs Guidance: 2023).

When deploying a surveillance camera system, you must have a clear understanding of your organisation’s responsibilities under data protection laws. The General Data Protection Regulation (GDPR or EU GDPR across Europe) and the Data Protection Act 2018 are formal laws backed up by heavy sanctions including significant fines and breaches that attract

negative media coverage. The Information Commissioner or equivalent offices in many other countries have published detailed guidance on how to create data protection impact assessments (DPIAs). If you are making decisions around capturing personal data captured as a controller, or joint controllers, you are responsible for compliance with data protection law, including the requirement to carry out a DPIA. It is recommended that DPIAs are carried out when:

- New systems are installed;
- Cameras are added or removed from systems;
- Cameras are moved or change position;
- Whole or parts of systems are upgraded; and
- Where systems that include biometrics capabilities such as automatic facial recognition are in use.

During and after cyber terrorism events, organisations remain legally responsible for data and privacy breaches within the UK, EU, USA and elsewhere. The fact that an organisation and its staff suffered a severe shock (read ‘disruption’ in business continuity parlance) does not reduce or remove this 24/7 legal obligation. Moreover, following major security incidents, organisations and individuals – including their electronic devices – caught up in the event are prone to attract unwanted attention from all manner of hackers. There is also a likelihood that communications data and all CCTV-related data could be seized or subpoenaed by investigative agencies, regulatory bodies and/or formal inquests. Therefore, if CCTV or other personally identifiable information (PII) is stored within Cloud services, do ensure that strong security settings including **multifactor authentication (MFA)**, firewalls, further backup solutions and decent encryption are consistently applied in and around your Cloud. Cyber attacks on organisational Cloud storage backup solutions, conducted by a mix of hostile state actors and pro-extremist hacktivist groups, are exponentially rising.

UK cyber crime laws

Computer Misuse Act (1990): According to the government, “the Computer Misuse Act 1990 (CMA) is the main legislation that criminalises unauthorised access to computer systems and data, and the damaging or destroying of these. The Act has the intention of protecting the integrity and security of computer systems and data through criminalising access to them which has not been authorised by the owner of the system or data” (Home Office: 2023a). In May 2021, the Home Secretary announced a review of the CMA because law enforcement and industry felt that malicious acts on computer systems, and areas necessary for effective incident response, were not adequately addressed by the current laws. In some scenarios, cyber investigators faced prosecution for unauthorised access into external computer systems. Proposed areas for legal enhancement include: providing extra legal powers for authorities or other agreed bodies to take down domain names and **IP addresses** used for clearly criminal activity, the takeover of IP and domain names, powers to ensure data preservation, unauthorised data copying becoming treated as ‘theft’, as well as new powers to launch cyber investigations overseas (Home Office: 2023a).

Data Protection Act (2018) – DPA: UK only. Replaces the Data Protection Act (1998) and incorporates the EU GDPR into British law. DPA installs the Information Commissioner’s Office (ICO) as the enforcement authority. The 2018 Act establishes the importance of organisations to be more *responsible* with personal data as well as to improve and safeguard data *confidentiality*. Suspected data breaches must be reported to the ICO within 72 hours and data subjects must be informed. DPA has seven parts, including the processing of data by competent bodies and the intelligence services, which falls beyond the GDPR.

One requirement under the DPA/GDPR is that an organisation undertakes a **Data Protection Impact Assessment (DPIA)**: According to the ICO, a DPIA is a process to help you identify and minimise the data protection risks of a project. You must do a DPIA for processing personal data that is likely to result in a high risk to individuals. Your DPIA must:

- Describe the nature, scope, context and purposes of the processing;
- Assess necessity, proportionality and compliance measures;
- Identify and assess risks to individuals; and
- Identify any additional measures to mitigate those risks.

A full list of all cyber crime and prosecutor’s guidance is produced by the Crown Prosecution Service (CPS). Some of these laws may be very useful for businesses and/or police forces that could conduct investigations, or seek to make other forms of early interventions, to prevent politically violent radicalisation or physical attacks. The online guide covers areas ranging from online harassment, unlawful eavesdropping and surveillance, economic crime and fraud, and cyber attacks, *inter alia*. The CPS’s tabularised summary of all criminal laws related to cyber crime is provided in this book’s Appendix E.

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US

The First Amendment of the United States Constitution determines that Congress makes no law respecting an establishment of religion or prohibiting its free exercise. It protects freedom of speech, the press, assembly, and the right to petition the government for a redress of grievances. Thus, the US body politic has been cautious to regulate any form of freedom of speech although the PATRIOT Act (described below) fundamentally shifted power into Federal government security agencies and national law-enforcement forces.

PATRIOT Act: The *Uniting and Strengthening America by Providing Appropriate Tools Required to Intercept and Obstruct Terrorism Act* became signed into law during 2001 by President George W. Bush after the 9/11 terrorist atrocities and subsequent anthrax attacks within the US. The legislation sought to bolster US national security, particularly against foreign-related acts of terrorism. The PATRIOT Act is expansive and wide-ranging, increasing surveillance (including phone-tapping), easing interagency communications and increasing sentences for terrorism-related offences.

For instance, Section 505 authorises government agencies to seize financial, Internet, credit and telephone records without prior judicial review and without articulable suspicion that the target is a terrorist or a spy. Another area that strengthened powers for ICT monitoring was published under Section 901. This section permits the US intelligence community head (CIA director) to set “*requirements and priorities*” for domestic intelligence-gathering, which could include domestic communications surveillance and monitoring (American Civil Liberties Union: 2023).

In 2012, President Barack Obama signed the PATRIOT Sunset Extensions Act into law, providing for roving wiretaps, business-record searches and conducting surveillance on lone wolves. According to the US Office of Justice Programmes: “*The Act also expands the list of situations in which the Secret Service may participate in the investigation of computer crimes and authorizes the U.S. Attorney General to establish regional computer forensic laboratories to train Federal, State, and local law enforcement in investigating computer crimes*”. More specifically for cyber-related concerns, including terrorism financing: “*the [PATRIOT] Act expands the scope of key money laundering statutes; includes computer fraud and abuse as part of the grounds for charges of terrorism; and modifies the computer fraud statute, including permitting extraterritorial jurisdiction*” (Office of Justice Programs: 2002).

US cyber crime laws

Computer Fraud and Abuse Act (CFAA): Passed in 1984 as an amendment to the rather weak Comprehensive Crime Control Act (1984). CFAA came from a strong concern that federal and other governmental networks were being targeted by skilled hackers. Specifically, the CFAA prohibits the following:

- Knowingly accessing a computer without authorisation to obtain national security or other government-restricted data.
- Intentionally accessing a computer without authorisation to obtain certain information from:
 - o A financial institution or consumer reporting agency;
 - o The federal government; or
 - o A protected computer.
- Intentionally accessing and affecting the use of a government computer.
- Knowingly accessing a protected computer to defraud and obtain anything of value.
- Causing damages specified in the statute by knowingly transmitting harmful items or intentionally accessing a protected computer.
- Knowingly trafficking in computer passwords.
- Extortion involving threats to damage a protected computer.

“*In certain circumstances,*” say legal experts, “*the CFAA permits an individual who suffers damages to bring a civil action for damages or injunctive relief against a violator*” (Thomson Reuters Practical Law website: 2023). The Act’s network security provisions (drafted pre-World Wide Web) have been amended and strengthened several times, including by the PATRIOT Act. In the wake of computer researcher Aaron Swartz’s suicide, some US legislators sought to amend and limit some CFAA provisions so that it would not inadvertently capture most white or grey hat hacking activities. Aaron Swartz, a computer programmer and Internet hacktivist, had been charged by federal prosecutors on two counts of wire fraud and eleven violations of the CFAA. Unhappy that his plea bargain for a lighter sentence had been rejected, Swartz took his own life in 2013.

Health Insurance Portability and Accountability Act (HIPAA): Signed into law by President Bill Clinton during August 1996. HIPAA modernised health data processing laws to protect the data subject and the transmission of healthcare information. It stipulates how personally identifiable information (PII) acquired, stored and disseminated by the healthcare and healthcare insurance industries should be protected from fraud and theft. Fines and sanctions find their way to organisations that are found to be responsible for lax security and breaches.

North American Electric Reliability Corporation Critical Infrastructure Protection (NERC CIP): This sector partnership was originally established during the 1960s out of concerns about preventing power blackouts in the US and Canada. Now it is very much engaged in assisting its members to prevent cyber terrorism. According to industrial power experts, Verve: *“Discussions around the consideration of the creation of a set of cyber security standards for the industry began when the catalyzing events of 9/11/2001 occurred and provided an increased sense of urgency to the effort”* (Verve: 2022). The NERC CIP standards are now mandatory security standards that apply to all entities that own or manage facilities forming part of the Canada and US electric power grid. These standards were initially approved by the Federal Energy Regulatory Commission (FERC) in 2008. According to TechTarget’s Rahul Awati and Ben Cole: *“These standards carry the force of regulations, meaning they are required by law [...] All entities that fall under the purview of NERC CIP must comply with these standards”* (TechTarget: 2023). The following graphic illustrates NERC CIP’s nine golden rules to safeguard cyber security:



Figure 1: NERC’s Nine Rules¹

Federal Information Security Modernization Act (FISMA):S. 2521. Signed into Federal law in December 2014 by President Barack Obama. Commonly referred to as FISMA Reform, this legislation was passed in response to an increasing amount of significantly impactful cyber attacks on federal government systems. FISMA changed existing laws to enable the federal government to more effectively respond to cyber attacks on departments and agencies.

Communications Decency Act: Following the January 2021 crowd attack on Congress in Washington DC, we may well be about to witness “*a new era of social media regulation*” argues *Harvard Business Review* columnist, Dipayan Ghosh (2021). In an incident that rocked America, and shocked much of the world, seven people were killed and 140 police officers injured. Afterwards, social media companies participated in a relatively synchronised takedown of around 70,000 QAnon conspiracy theory group members’ social media accounts. Tech firms moved swiftly and controversially. Donald Trump’s Twitter (now X) feed was shut down and Snapchat imposed a temporary lockout for the former president. In addition, YouTube removed what it considered to be a handful of community guideline-breaching videos, citing their inflammatory nature. Stripe’s online payment platform ceased processing Trump campaign website payments.

Since this historic incident, a groundswell of senior figures across US government and both Houses of Congress have moved to strengthen or remove Section 230 of the Communications Decency Act (CDA 1996). This is the federal law provision that grants Internet service providers protection from liability (aka immunity) for user-generated content, hosted and disseminated on their platforms (Ghosh: 2021). In the US and across Europe, ISPs including Google, Twitter, AOL, Grindr and Facebook enjoyed their legal immunity being upheld, seeing cases against them dismissed, although their servers and services carried undoubtedly inflammatory, dangerous and brazenly false or conspiratorial communications.

The *Force v. Facebook Inc.*, 934 F.3d 53 (2nd Cir. 2019) second appeals court decision held that Section 230 bars civil terrorism claims against social media companies and Internet service providers despite expert testimony asserting that: “*Mounting evidence suggests that providers designed their algorithms to drive users toward content and people the users agreed with — and that they have done it too well, nudging susceptible souls ever further down dark paths*” (McCabe: 2021). The case against Facebook had been brought under the US Anti-Terrorism Act by families and friends of several killed by terrorists in Israel. Hamas had reportedly used the social media platform to coordinate activities.

(Proposed) Social Media Child Protection Act: At the time of writing, the US Congress 2023–24 session were discussing passing legislation “*To require providers of social media platforms to prohibit children under the age of 16 from accessing such social media platforms, and for other purposes*” ([Congress.gov](https://www.congress.gov): 2023). If passed, the Social Media Child Protection Act will very much supersede existing provisions under the Children Online Privacy Protection Act (1998) that curb marketing directly to minors and acquiring children’s personal data without parental consent.

Europe and EU

In major European economies, successful cases have been brought – albeit rarely – against directors of tech companies whose systems and platforms have published words and/or images that breach criminal or libel-related laws. Although in these exceptional cases costly appeals have sometimes been won, there is a clear political/governance trajectory moving towards more regulation and liability placed on the shoulders of ISPs and social media platforms. One successful appeal was brought by business leader, Felix Somm, former managing director of CompuServe Germany. Somm became charged with violating child pornography laws because of content being carried on CompuServe’s servers stored within Germany. In the initial criminal case (1998), Somm was convicted and sentenced to two years’ probation (Kuner 1998). Somm was cleared on appeal the following year. In the original case, the judge appeared unimpressed by the technical defence that content filtering was impossible when he declared that CompuServe had let “*protecting the young ... take second place to maximizing profits*” (Internet Archive Wayback Machine: 2023). Legal interpretations can appear fickle with devastating consequences for business owners and security managers that are slow to address potential criminal activities hosted on their networks.

EU General Data Protection Regulation (EU GDPR): Adopted by the EU in 2016 and enforceable since 2018. A key component to the GDPR is that informed consent must have been acquired by the user from the data subject, for the subject’s data to be used. The regulation applies if the data controller (an organisation that collects information about living people, whether they are in the EU or not), or processor (an organisation that processes data on behalf of a data controller like Cloud service providers), or the data subject (person) is based in the EU. Under certain circumstances, the regulation also applies to organisations based outside the EU if they collect or process personal data of individuals located inside the EU. The regulation does not apply to the processing of data by a person for a “*purely personal or household activity and thus with no connection to a professional or commercial activity*”. Similar or mirror legislation has been passed in several non-EU states and territories including the UK, California (The California Privacy Rights Act (CPRA), Turkey, Japan, Kenya, Brazil, Argentina and South Korea.

This document was truncated here because it was created in the Evaluation Mode.